

HEDGE-05 Null Result

HEDGE-05 NULL RESULT — null_cost

i Firing Condition

This null-result PDF was generated because firing condition **null_cost** tripped.

Evidence Block

Evidence: Phase-0 Cost-Leg Gate Failure (null_cost)

The free-tier subgraph cost-leg lower-bound check (REPRO-03) failed against the PRE_REGISTRATION-locked threshold. See `notes/<protocol>_cost_leg_bound.md` for the verdict + threshold + measured query rate.

Per HEDGE-05 firing condition (a), no convex-hedge claim is justified for the candidate substrate at this iteration; the candidate is RETIRED-PENDING-MATURITY.

Cost-leg verdict (the gating evidence)

Field	Value
Verdict	FAIL
Firing condition	null_cost
Demand band (queries/mo)	30000–100000
Free-tier ceiling (DEMAND-01)	100000
Pre-registered rule (AF-03)	not strictly > 100k -> FAIL (pre-registered, AF-03)

The estimated demand **straddles** the free-tier ceiling (not strictly above it), so the pre-registered conservative-fail rule fires **null_cost**.

DGP support (produced by the full re-run; not the gating verdict)

The frozen Phase 2-5 pipeline ran in full regardless of the cost-leg fail (that is what demonstrates REPRO-02). Its output is preserved as supporting context – it does NOT override the cost-leg gate:

Diagnostic	Value
Branching ratio $\hat{\eta}$	0.709
Branching-ratio CI	[0.001, 0.95] (width 0.949)
LR test (NHPP null)	p = 0.0
Convex-dominance: vol-of-vol > 0	{‘passed’: False, ‘evidence’: {‘per_leg’: {‘leg_0’: {‘vol_of_vol’: 0.4329166177597581, ‘rolling_var_mean’: 0.7988787037569063, ‘rolling_var_std’: 0.3458478664307396, ‘window’: 10}, ‘leg_1’: {‘vol_of_vol’: 0.5426796662456415, ‘rolling_var_mean’: 0.42018848493494976, ‘rolling_var_std’: 0.22802774676476026, ‘window’: 10}}, ‘threshold’: 0.6}}}
Convex-dominance: positive skew + fat tails	{‘passed’: False, ‘evidence’: {‘per_leg’: {‘leg_0’: {‘skew’: nan, ‘excess_kurt’: nan}, ‘leg_1’: {‘skew’: nan, ‘excess_kurt’: nan}}, ‘kurtosis_excess_threshold’: 0.0}}}
Convex-dominance: Hawkes self-excitation	{‘passed’: True, ‘evidence’: {‘branching_ratio’: 0.7088622072760358, ‘branching_ratio_floor’: 0.2, ‘eta_floor_met’: True, ‘adjacency_off_diag_max’: 0.0003138942591323242, ‘canonical_ll_source’: ‘abrigo_x402.dgp.lr_test._hawkes_loglik_vectorized’}}
Convex-dominance: USDT-depeg basis jump	{‘passed’: True, ‘evidence’: {‘source’: ‘literature_range_stipulation’, ‘base_triple’: {‘lambda_J’: 0.05, ‘mu_J’: -0.05, ‘sigma_J’: 0.02}, ‘sensitivity_fragile’: False, ‘sensitivity_summary’: {‘n_samples’: 64, ‘n_flips’: 0, ‘flip_examples’: []}}}

Diagnostic	Value
Any convex-dominance condition passed	True
Q9 trigger (CI width > 0.4)	True (V3-anchor-only; unified fallback pre-registered DEFERRED)

REPRO-02 attestation

This deliverable was produced by re-running the **frozen** Phase 2-5 pipeline on a different protocol via a **config-swap only** (`protocols/*.toml`): the `git diff` over `fetch/src` + `analysis/src` across the iteration boundary is **empty**. The swap-surface invariant (REPRO-01/REPRO-02) holds; the `null_cost` verdict is recorded **as observed** (the FEATURES.md D-08 negative control).

Pre-Registration Reference

This null result is a valid completion per [PRE_REGISTRATION.md §Decision Rules](#). Null results are explicitly publishable per AF-03 discipline.